

WASP-177b

R-2448

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-177_230803 · created 2026-08-22T23:51:31+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460159.94 +/- 0.01 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.138 +/- 0.049
Transit depth (Rp/Rs)^2	1.89 +/- 1.34 [%]
Orbital Inclination (inc)	89.13 +/- 2.94
Airmass coefficient 1 (a1)	1.012 +/- 0.038
Airmass coefficient 2 (a2)	0.004 +/- 0.044
Scatter in the residuals of the lightcurve fit is	3.72 %
Best Comparison Star	#2 - [341, 293]
Optimal Aperture	2.38
Optimal Annulus	5.83
Transit Duration (day)	0.099 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.138 ± 0.049 vs 0.1618 (deviation 0.47σ)
Duration fit vs published	0.099 d vs 0.0721 d (deviation 1.3σ)
Mid-time O–C	7.6 min
Depth SNR	2.7
Residual scatter	3.72 %

Light curve

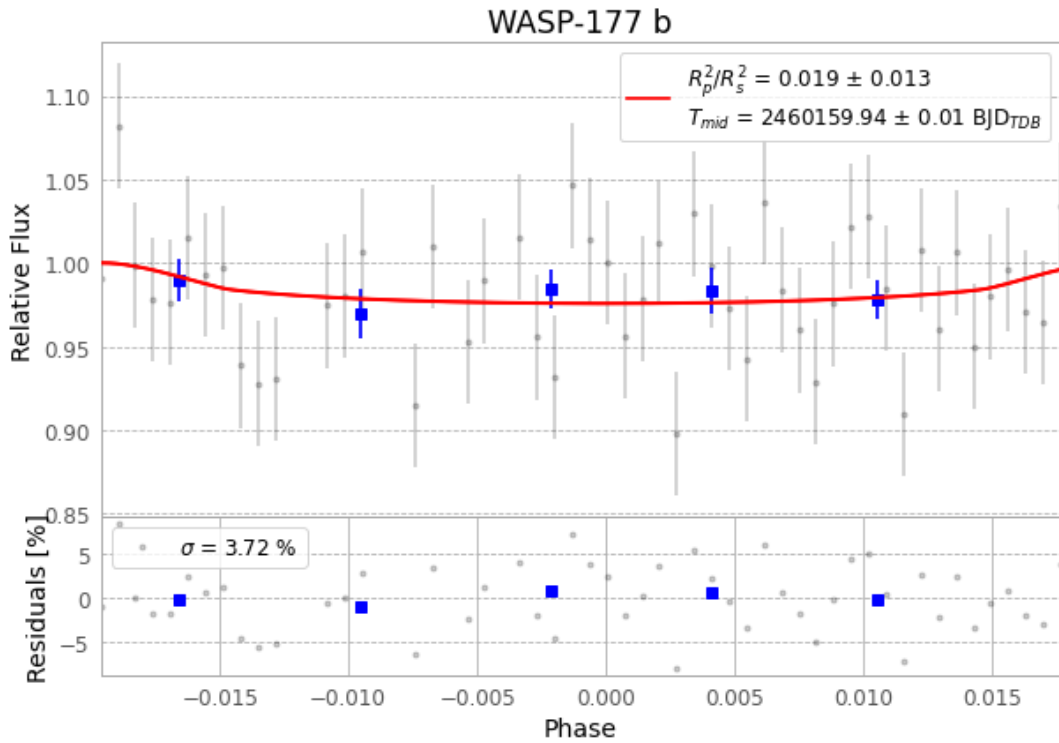


Figure 1 — FinalLightCurve_WASP-177 b_2023-08-03.png (SHA-256 607963a3882f4806...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [259, 179], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d1a683d47f7d81d9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

56 FITS frames (37 MB), WASP-177230803085715.FITS → WASP-177230803114215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-177-230803.log (059bdcf2e14f...), FinalLightCurve_WASP-177 b_2023-08-03.png (607963a3882f...), FinalLightCurve_WASP-177 b_2023-08-03.csv (239117402c43...)

Compute resources

Wall time 354.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-22 23:51Z.